

PRE-LAB 6: Pointers, Arrays, and Functions

This pre-lab helps you practice pointers, arrays, and functions in C. The problems are simple and inspired by everyday scenarios. Each question includes a solution.

Question 1: Average Steps

Description: A person recorded the number of steps they walked for 5 days. Write a function that calculates the average steps.

Function Template:

```
int average_steps(int steps[], int size);
```

Example:

```
Input: {8000, 9000, 10000, 7000, 8500}  
Output: 8900
```

Solution:

```
#include <stdio.h>  
  
int average_steps(int steps[], int size) {  
    int sum = 0;  
    for(int i = 0; i < size; i++) {  
        sum += steps[i];  
    }  
    return sum / size;  
}  
  
int main() {  
    int steps[] = {8000, 9000, 10000, 7000, 8500};  
    int avg = average_steps(steps, 5);  
    printf("Average steps: %d\n", avg);  
    return 0;  
}
```

Question 2: Weekly Temperature Chart

Description: You have weekly temperature data stored in an array. Write a function to print a temperature chart using *, where each * represents 1 degree.

Function Template:

```
void print_temperature_chart(int temps[], int size);
```

Example:

```
Input: {3, 5, 2}
Output:
```c
Day 1: ***
Day 2: *****
Day 3: **
```

**Solution:**

```
#include <stdio.h>

void print_temperature_chart(int temps[], int size) {
 for(int i = 0; i < size; i++) {
 printf("Day %d: ", i + 1);
 for(int j = 0; j < temps[i]; j++) {
 printf("*");
 }
 printf("\n");
 }
}

int main() {
 int temps[] = {3, 5, 2};
 print_temperature_chart(temps, 3);
 return 0;
}
```

**Question 3: Total Income from Daily Sales**

**Description:** Given daily sales amounts, write a function that returns the total positive sales (ignore returns, i.e., negative numbers).

**Function Template:**

```
int total_positive_sales(int sales[], int size);
```

**Example:**

Input: {100, -20, 50, -10}  
Output: 150

**Solution:**

```
#include <stdio.h>

int total_positive_sales(int sales[], int size) {
 int sum = 0;
 for(int i = 0; i < size; i++) {
 if(sales[i] > 0) sum += sales[i];
 }
 return sum;
}

int main() {
 int sales[] = {100, -20, 50, -10};
 int total = total_positive_sales(sales, 4);
 printf("Total positive sales: %d\n", total);
 return 0;
}
```

**Question 4: Reverse an Array of Scores**

**Description:** Reverse an array of scores in place using a loop.

**Function Template:**

```
void reverse_array(int arr[], int size);
```

**Example:**

```
Example:
```c
Input: {10, 20, 30, 40}
Output: {40, 30, 20, 10}
```

Solution:

Solution:

```
#include <stdio.h>

void reverse_array(int arr[], int size) {
    for(int i = 0; i < size/2; i++) {
        int temp = arr[i];
        arr[i] = arr[size-1-i];
        arr[size-1-i] = temp;
    }
}

int main() {
    int scores[] = {10, 20, 30, 40};
    reverse_array(scores, 4);
    printf("Reversed scores: ");
    for(int i = 0; i < 4; i++) {
        printf("%d ", scores[i]);
    }
    printf("\n");
    return 0;
}
```